Mohamed Fedi BOUABID

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EDUCATION

National Institution of Applied Sciences and Technology **Sep. 2020 - Sep. 2025**
Engineering degree, Data science and intelligent systems *Tunis, Tunisia*

- **Relevant coursework :** Algebra, Calculus, Probability and Statistics, Algorithm, Optimization, fuzzy logic, machine/deep learning, big data.

WORK EXPERIENCE

EFREI Research Lab **Aug. 2023 – Aug. 2025**
NLP and Graph R&D Intern *Villejuif, France*

- Engineered an end-to-end pipeline for collecting, cleaning, and structuring multi-platform social media data (1M+ posts),reconstructing conversation trees and user interaction graphs with NetworkX.
- Implemented and benchmarked text classification models, fine-tuning a BERTweetFr model achieve a 94.2% F1 score for domain-specific relevance detection.
- Integrated NLP outputs with graph structures for semantic enrichment, analyzing the correlation between sentiment, topic relevance, and network motifs to explain user engagement patterns.
    - **Stack: NetworkX, PyTorch, HuggingFace, Transformers, Cytoscape, BERT, Gemini.**

E-Direct Tunisie **Jul. 2024 – Aug. 2024**
AI Developer Intern *Tunis, Tunisia*

- Designed and developed a web platform for automated recruitment.
- Integrated AI models for extracting, analyzing, and evaluating applications.
- Wrote comprehensive technical documentation to ensure the solution's maintainability and scalability
    - **Stack : Angular, NestJS, Flask, MySQL, YOLOv8, Gemini 2.0, OCR, LaTeX.**

La Poste Tunisienne **Jul. 2023 – Jul. 2023**
Fullstack Developer Intern *Ben Guerdane, Tunisia*

- Developed a centralized platform to manage office inventory and automate the supply process, reducing human errors by 40% and cutting request processing times by 90%.
    - **Stack : NestJS, Angular, MySQL, JWT.**

PROJECTS

Job Email Classifier Dashboard | Github repository **2025**

- Streamlit dashboard that integrates Gmail and Gemini AI to automatically classify and visualize job-related emails with interactive analytics helping job seekers manage email responses.
    - **Stack : Langchain, Gemini API, Gmail API, Streamlit.**

NutriWise **2024**

- Mobile App that uses ML models hosted on Azure ML platform to assess food healthiness based on nutritional values retrieved from barcodes.
    - **Stack : Flutter, Firebase, Sci-kit learn , google ML Kit, Azure machine learning, Flask.**

RAG Project | Github repository **2024**

- Streamlit-based application that enables conversational querying of multiple PDF documents with natural language and interactive chat history.
    - **Stack : Langchain, ChromaDB, Llama 3.1, nomic-embed-text, Streamlit.**

MLOps Project | Github repository **2024**

- Hosted an XGBoost model on a Flask server within Docker containers, implemented monitoring with Prometheus and Grafana, and migrated the entire setup to Azure Machine Learning for scalable deployment.
    - **Stack : Sci-kit learn, Prometheus, Grafana, Docker, Azure ML.**

SKILLS

- **AI & Data Science :** NLP (Transformers, Hugging Face, PyTorch) · Topic Modeling (LDA, BERTopic) · Computer Vision (OCR, Vision Transformers) · RAG & AI Agents · Scikit-Learn · Data Analysis (Pandas)
- **Full-Stack :** Angular · NestJS · Flask · Django · JWT Authentication · MySQL · PostgreSQL
- **Cloud & Tools :** Firebase · Azure · Streamlit · Flutter · Power BI · Git · LaTeX
- **Languages :** French (Advanced Intermediate B2), English (Advanced Intermediate B2), Arabic (Native)